

Phase 2 Results: Topology, Identity Mix, and Scoring Mode in a Climate-Firestorm Simulation

TUM M.Sc. Data and Society — thesis results chapter (standalone)

Harvest stamp: 19 September 2026

This chapter reports the Phase 2 campaign of the climate-firestorm simulation. The campaign crossed eight network topologies with homogeneous (H) and heterogeneous (He) identity mixes and with two auditor instruments, then compared a reconstructed Debnath hashtag graph against a custom-graph (D-net) simulation. Discrete MPR (dual-run headline) and continuous MPR (continuous-only headline) are different instruments; this chapter never averages them. The harvest is a real `gpt-4o-mini` run with $N = 1$, eight hops, and `thesisGrade=false`. Hatched dead cells remain in the grid. The Debnath comparison is structural: 814k tweet identifiers were not hydrated, and Debnath’s skip-gram window of eight is not this protocol’s hop budget.

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1 Purpose of this chapter

This chapter reports the Phase 2 results of the climate-firestorm simulation. The purpose is to describe, under documented limits, how eight network topologies, homogeneous versus heterogeneous identity mixes, and two auditor instruments move misinformation scores on six geoengineering articles. The chapter is a thesis-style results account. It is not a reprint of the CIKM linear-chain study on commercial and crime news (Maurya et al., 2025), and it is not a Twitter digital twin of Debnath and colleagues’ 814,924-tweet observational map (Debnath et al., 2023b).

The theoretical spine remains online firestorms as defined by Pfeffer, Zorbach, and Carley (2014): a sudden discharge of large quantities of negative word-of-mouth in social media, often with high affective charge. Their Outlook section lists *seven* interrelated factors, not six. Section 8 restates those seven factors from the Debnath-versus-D-net mapping and records the thesis six-knob list as a computational remapping. Cross-media dynamics has no engine knob and is held.

Phase 2 asked a narrower empirical question than a full Pfeffer confirmation. Holding surprise as a drip seed and holding the clock at eight ticks, do topology, identity mix, and auditor mode change live-cell mean MI on climate seeds? The harvest answers that question only descriptively. `summary.json` sets `thesisGrade` to false. $N = 1$ with graph seed 42 forbids error bars and replicate inference.

The remainder of the chapter follows a single path. Section 2 snapshots the protocol. Section 3 separates discrete from continuous MPR. Section 4 reports all eight topologies. Section 5 reports personas and mixes without averaging the two instruments. Section 6 reports the five numbered comparisons from the full analysis (C1–C5). Section 7 states that cell pooling concatenates rows and is not a physical super-graph. Section 8 compares D-net structure and Pfeffer observables with the Debnath hashtag fallback. Section 9 states limitations. Section 10 closes.

Figures are copies of `thesisExperiment/analysis_full/figures/` (fig00–fig22). Comparison panels follow `COMPARISONS.md`. Exploratory rank tests in that file are not treated as replicate inference.

2 Methods snapshot

The Phase 2 campaign reused the existing CIKM engine entry point (`node index.js -config`) with thesis-layer configs under `thesisExperiment/configs/phase2/`. Each node is a persona-conditioned `gpt-4o-mini` agent. Allowed actions are forward, reinterpret, and drop. Propagation runs first; the auditor scores rewrites in a second pass (Maurya et al., 2025). The auditor is also `gpt-4o-mini`, so the scores are LLM-as-judge values in the sense of Zheng et al. (2023), not human ratings.

Figure 1 sketches the two-phase pipeline. A ground-truth article enters at the drip seed. Persona-conditioned nodes rewrite or forward the text along the graph. The auditor returns an event MI on five yes/no items. Discrete MI equals the count of missing plus incorrect answers and therefore sits in $\{0, 1, 2, 3, 4, 5\}$. Continuous MI is a float on approximately the same 0–5 range. Node MPR is the mean of scored event MI for that node and article. The cell headline used

Phase 2 climate-firestorm loop (CIKM grammar: MI/MPR/severity; new grid)

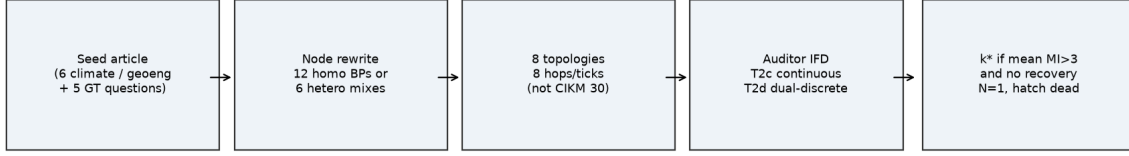


Figure 1: Methods schematic for Phase 2. A ground-truth climate article is rewritten by persona-conditioned `gpt-4o-mini` nodes on an eight-node graph, then scored by a five-item auditor. Discrete and continuous campaigns are separate configs. Network k^* is the first irreversible crossing of mean $MI > 3$, censored at eight ticks. Source: `analysis_full/figures/fig00_methods.png`.

Table 1: Phase 2 harvest grid. Live means follow `analysis_full/COMPARISONS.md`. Dead cells are hatched after real LLM calls (`nScored ≤ 1, llmCalls > 0`) and are excluded from live means.

Slice	Cells	n live	Hatched dead	meanMI (live)
T2c_H (cont., homo.)	576	480	96	0.8643
T2d_H (dual-disc., homo.)	576	478	98	1.6814
T2c_He (cont., hetero.)	288	238	50	1.0192
T2d_He (dual-disc., hetero.)	288	240	48	2.0800
D-net custom graph	8 article-rows	8	0	(separate graph)
Thesis grid (T2 only)	1,728	1,436	292	<code>thesisGrade</code> false

An earlier harvest brief counted 290 hatched rows. The full-analysis tables count 292. This chapter uses 292. D-net is not concatenated into the 1,728-cell grid.

below is parser `meanMI` (mean scored-event MI on live cells with `nScored > 1`). `meanNodeMPR` is the mean of node means of the same headline. The two scalars sit beside each other; they are not mixed into one number.

2.1 Grid that was harvested

Phase 2 isolated its outputs from Phase 1. Runs live under `thesisExperiment/runs_phase2/`; tables live under `thesisExperiment/results_phase2/`. The discrete 12×12 Phase 1 campaign in `runs/` and `results/tables/` was not overwritten and is not spliced into the headlines below.

Table 1 records the harvested grid. T2c denotes continuous-only scoring. T2d denotes dual scoring, whose *headline* is discrete. Homogeneous (H) cells cycle one of twelve personas on eight nodes. Heterogeneous (He) cells cycle one of six mixes of eight persona identifiers. Six core articles run sequentially in each config: SCoPEX 2017, chemtrails–Gates 2018–2021, SAI geoengineering, the Paris Agreement, climate consensus, and polar bears. The expected thesis grid is 1,728 cells. The harvest reports 0 missing cells.

2.2 Shared protocol

Every T2 cell used eight nodes, eight ticks, eight hops, inbox cap 4, trust threshold 0.15, action weights 0.35/0.50/0.15 (forward/reinterpret/drop), relation evolution on (`trustDelta`= 0.05), activity pattern `always`, and `graphRandomSeed`: 42. The seed is always `node_0` (drip). The

model is `gpt-4o-mini` for agent and auditor. Replicates: $N = 1$.

Eight hops is a logged cost cut versus the CIKM chain length of 30 (Maurya et al., 2025). It is not Debnath’s hop protocol. Debnath and colleagues define a skip-gram *context window of eight words* inside a tweet when they estimate p_{together} with “chemtrails” (Debnath et al., 2023b). That window is a word-embedding hyperparameter. It is not this campaign’s tick budget. The public R codes use an $n = 10$ n-gram implementation detail (Debnath, 2023); the paper text says eight.

2.3 Topologies and identity

The eight generators are `linear_chain`, `ring`, `random_er` (Erdős and Rényi, 1960), `small_world` (Watts and Strogatz, 1998), `scale_free` (Barabási and Albert, 1999), `echo_chamber`, `polarized`, and `hierarchical`. Linear chain is the CIKM object (Maurya et al., 2025). It is included so the climate corpus can be read against that topology, not because a chain is a firestorm clustering contrast. Scale-free uses Barabási–Albert $m = 2$. ER uses $p = 0.42$ with `minSeedOutDegree=2` so the drip seed cannot have out-degree zero.

Twelve homogeneous personas fall into three families: conspiracy (four), climate action (three), and science/environment (five). Belief profiles are theory-faithful reductions of Debnath discourse types, not HDBSCAN centroids on 814k tweets (Debnath et al., 2023b). Six heterogeneous mixes vary conspiracy share from four of eight (`mix_00`) to zero of eight (`mix_02`). `mix_02` is the expert-inclusive mix: climate-action, science, and environment identifiers only.

2.4 Pfeffer knobs: swept, held, measured

Pfeffer, Zorbach, and Carley (2014) do not name valence, surprise, identity, clustering, echo, and temporal acceleration as a canonical six-factor theory. The campaign treats those six names as engine knobs. Valence is varied by the six articles and by persona tone. Identity alignment is swept (H versus He). Network clustering is swept (eight topologies). Information echo is measured (homophily on the last-tick graph) and additionally wired in the echo-chamber and polarised generators. Surprise is held as drip seeding. Temporal acceleration is held at eight ticks. Cross-media dynamics, the sixth Outlook factor in 2014, has no broadcast agent and is held.

2.5 Dead cells and k^*

A cell is hatched dead when `nScored` ≤ 1 after real LLM calls. Dead cells are kept. They are not imputed as `MI=0` and they are not read as “mix immunises”. Live means drop those cells and report `n_live`.

Network k^* is the first tick at which network-mean `MI` > 3 (propaganda) and mean `MI` stays > 3 on every later tick with data. If the series later returns to ≤ 3 , k^* is undefined (none), not zero. The crossing is right-censored at eight ticks, so irreversibility is weak. Dual k^* uses the discrete series. Continuous k^* uses the continuous series. The two rates are not added.

Independent-cascade models treat adoption as a binary infection (Kempe, Kleinberg, and Tardos, 2003). This engine carries mutating text plus an auditor, so k^* clocks semantic degradation rather than a binary infected bit. The clock is still a simulation artefact on eight ticks, not an empirical firestorm duration.

Table 2: Live-cell headline MPR by arm and instrument. Dead cells excluded. $N = 1$. Do not pool the two columns or the two rows.

Arm	Cont. MPR (T2c)	n live cont.	Dual-disc. MPR (T2d)	n live dual
H (homogeneous)	0.8643	480	1.6814	478
He (heterogeneous)	1.0192	238	2.0800	240
He – H (same mode only)	0.1549	—	0.3986	—
Mean dual gap (sidecar)	—	—	0.8804 (H) / 1.1145 (He)	—

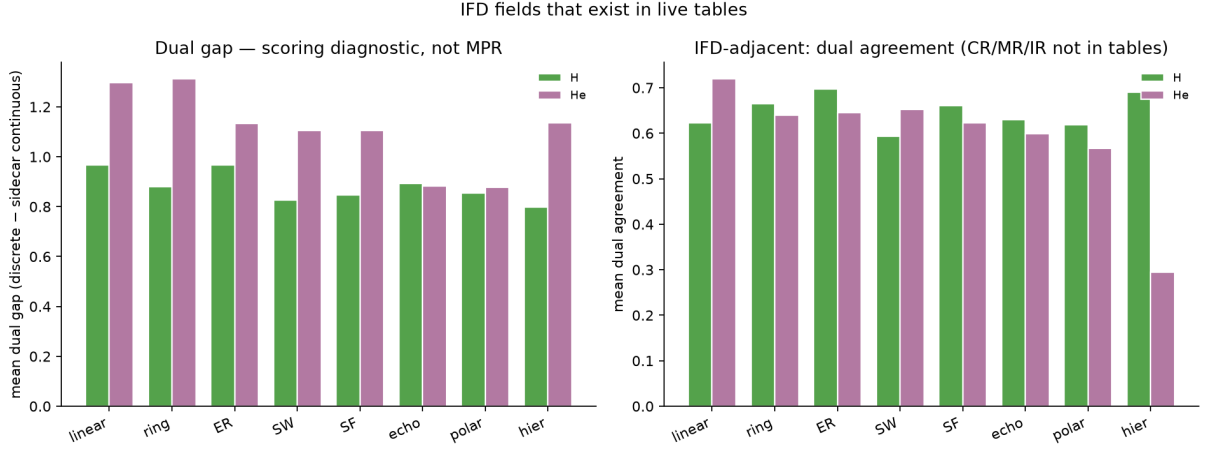


Figure 2: Dual gap (discrete headline minus sidecar continuous) and dual agreement. The gap is a scoring diagnostic, not a ground-truth MPR. $N = 1$, eight hops. Source: `fig20_ifd_dual_gap_agreement.png`.

3 Two scoring instruments

Phase 2 ran two auditor modes as separate campaigns. Continuous configs (prefix T2c) set `miScoringMode` to continuous. The headline MI/MPR is the continuous auditor float. Dual configs (prefix T2d) call both a discrete auditor and a continuous sidecar on the same events. The dual *headline* is the discrete 0–5 score. The sidecar continuous value is a scoring diagnostic. Dual gap (discrete minus sidecar continuous) is not a third ground-truth MPR.

These two headlines are not the same quantity. Dual-discrete H (1.6814) is not “the same MPR as continuous H (0.8643) but higher”. The auditors are different instruments. Dual-discrete sits higher than continuous on every topology in this harvest. That pattern is a scoring-mode effect. It is not evidence that the discrete campaign found more misinformation in a shared unit.

Table 2 reports live-cell means of `meanMI`. Dead cells are excluded. H and He remain separate. The harvest cross-check on H matches the recomputed live means to four decimals.

The same cells as `meanNodeMPR` (not interchangeable with `meanMI`) are H continuous 0.8630, H dual 1.6839, He continuous 1.0140, He dual 2.0737. This chapter leads with `meanMI` and treats `meanNodeMPR` as a sidecar of the same instrument.

Figure 2 plots dual gap and agreement. The gap is large on both arms. Using it as an MPR would smuggle the discrete–continuous disagreement into a third league table. The figure is a diagnostic only.

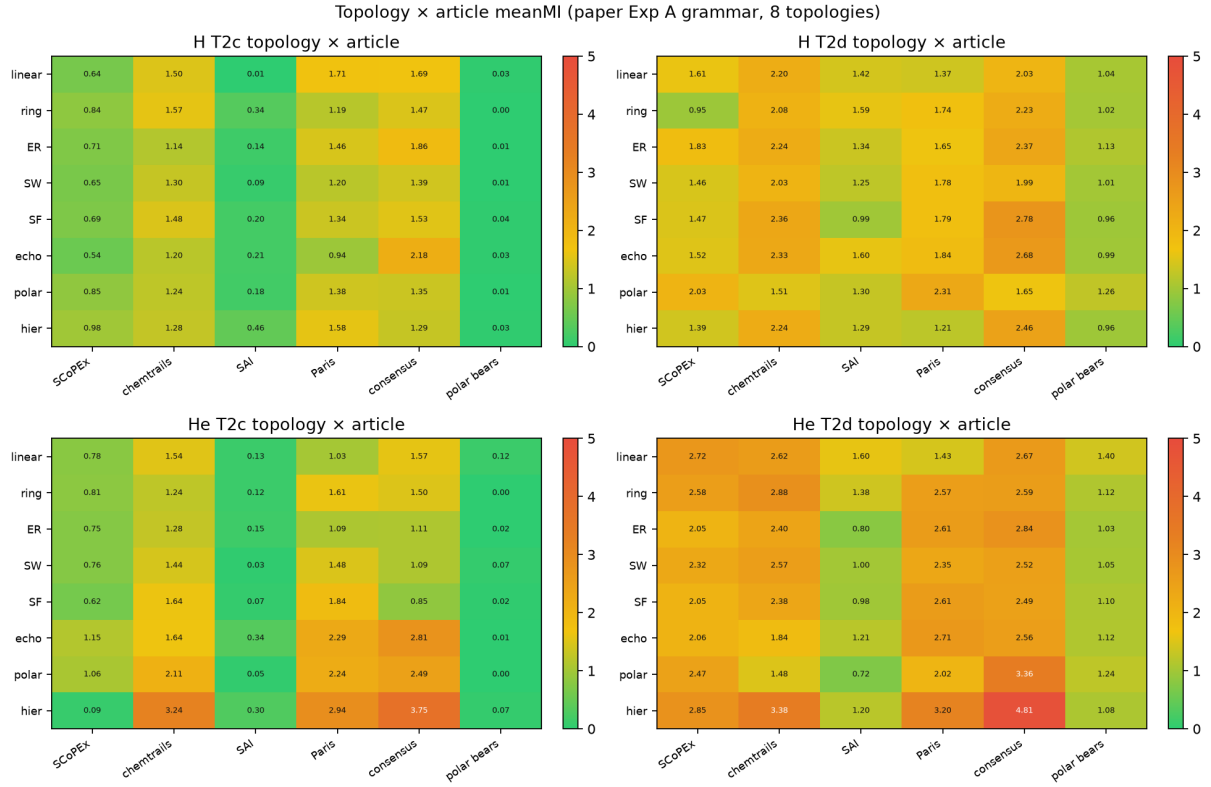


Figure 3: Live-cell mean meanMI by topology and article, all arms and modes. Discrete (T2d) and continuous (T2c) panels are not averaged. $N = 1$. Source: `fig17_topology_article_mpr.png`.

4 All eight topologies

Topology is the clustering knob in the remapping of Pfeffer, Zorbach, and Carley (2014). Phase 2 swept eight generators rather than the Phase 1 pair (scale-free versus ER). Figure 3 shows live-cell mean meanMI by topology and article across arms and modes. Rank order is not the same across scoring modes. A single “best topology” from a pooled number is not available from this harvest.

Tables 3 and 4 list the same means with live counts, k^* rates, and dead rates. Linear chain and ring are short, low-degree paths. Hierarchical is a different degree sequence. With $N = 1$ the gaps are descriptive. They are not a significance claim about ER versus scale-free.

On homogeneous continuous cells, scale-free is highest (0.9205) and small-world is lowest (0.7215). Echo-chamber H continuous is 0.8490 ($n = 63$ live). On homogeneous dual-discrete cells, echo chamber is highest (1.7984) and small-world is lowest (1.5977). On heterogeneous continuous cells, hierarchical is highest (1.5242) and ER is lowest (0.6593). On heterogeneous dual-discrete cells, hierarchical is highest (2.5934) and polarised is lowest (1.8830).

Hierarchical and polarised He cells also show high dead rates (about 28–31% continuous). A live-only mean on a half-dead topology inflates uncertainty. Those rows describe the surviving cascades, not a clean generator effect.

Figure 4 reports k^* occurrence on live cells. He dual-discrete k^* rates sit well above He continuous rates (32.1% versus 7.6% collapsed). The discrete instrument crosses and stays above 3 more often in this harvest. That is still a scoring-mode fact, not a second firestorm clock on a shared unit.

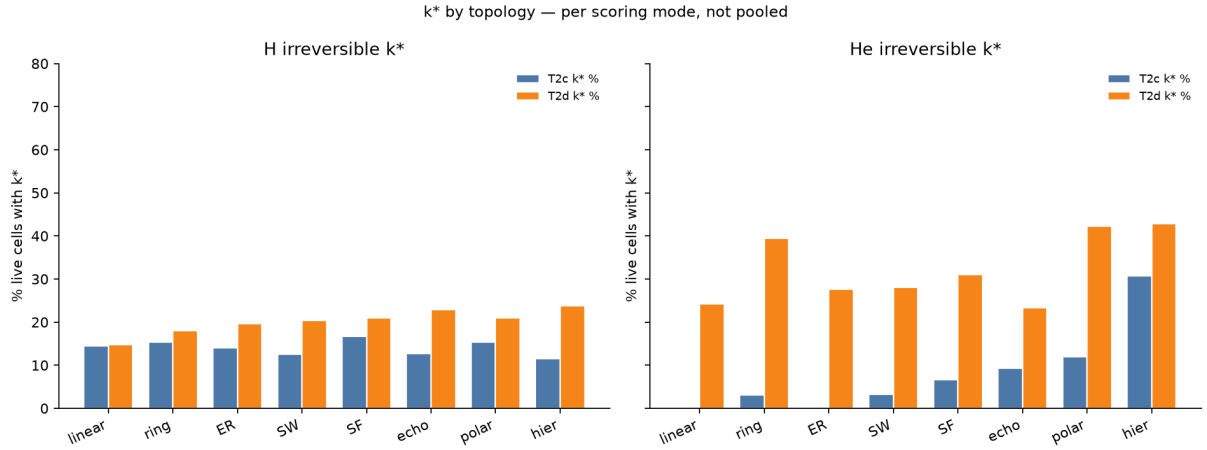


Figure 4: Share of live cells with irreversible network k^* (mean $MI > 3$ with no later recovery), by topology and scoring mode. Dual k^* uses the discrete series; continuous k^* uses the continuous series. The two rates are not added. Right-censored at eight ticks. $N = 1$. Source: `fig19_kstar_rates.png`.

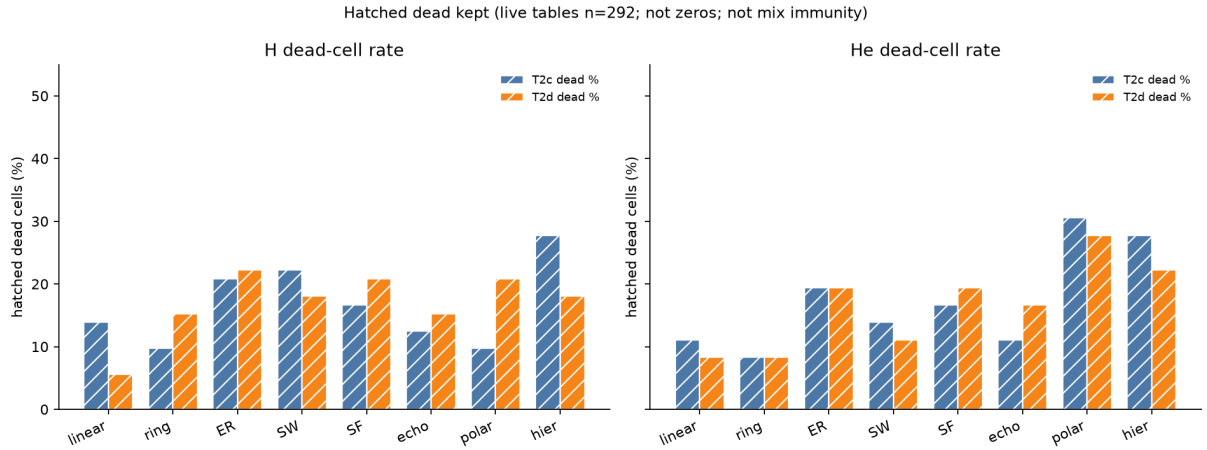


Figure 5: Hatched dead-cell rate by topology, arm, and scoring mode. Dead means $nScored \leq 1$ after real LLM calls. Hatched cells are kept and are not scored as mix immunity. $N = 1$. Source: `fig18_dead_rates.png`.

Figure 5 shows hatched dead-cell rates. Overall, 292 of 1,728 thesis cells are hatched after LLM. Hierarchical H continuous is 27.8% dead; polarised He continuous is 30.6% dead. Those generators contribute fewer scored events. The hatch pattern is part of the result, not a footnote.

5 Homogeneous versus heterogeneous identity

Identity alignment is the remapping of Pfeffer factor 5 (lack of diversity) into a persona mix (Pfeffer, Zorbach, and Carley, 2014). H cells use one persona on all eight nodes. He cells use a documented eight-identifier mix. Figure 6 plots collapsed live-cell means for each instrument.

Collapsed He is higher than collapsed H on both instruments (continuous +0.1549; dual-discrete +0.3986). That ordering is the opposite of a pooled “hetero buffers firestorms” claim of the sort associated with mixed CIKM chains (Maurya et al., 2025). The H mean mixes conspiracy personas (continuous about 2–2.5) with scientists (continuous near 0). The He mean mixes conspiracy-heavy `mix_00/mix_01` with conspiracy-free `mix_02`. Composition, not the H/He label, is what moves MPR in this harvest.

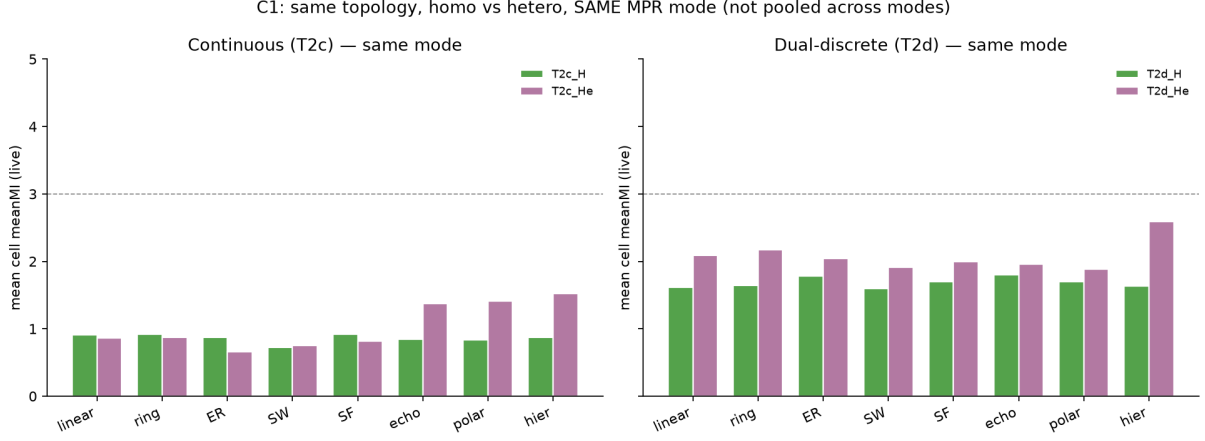


Figure 6: Same-mode H versus He by topology (C1). The two instruments are not averaged. $N = 1$. Source: `fig01_c1_h_vs_he_same_mode.png`.

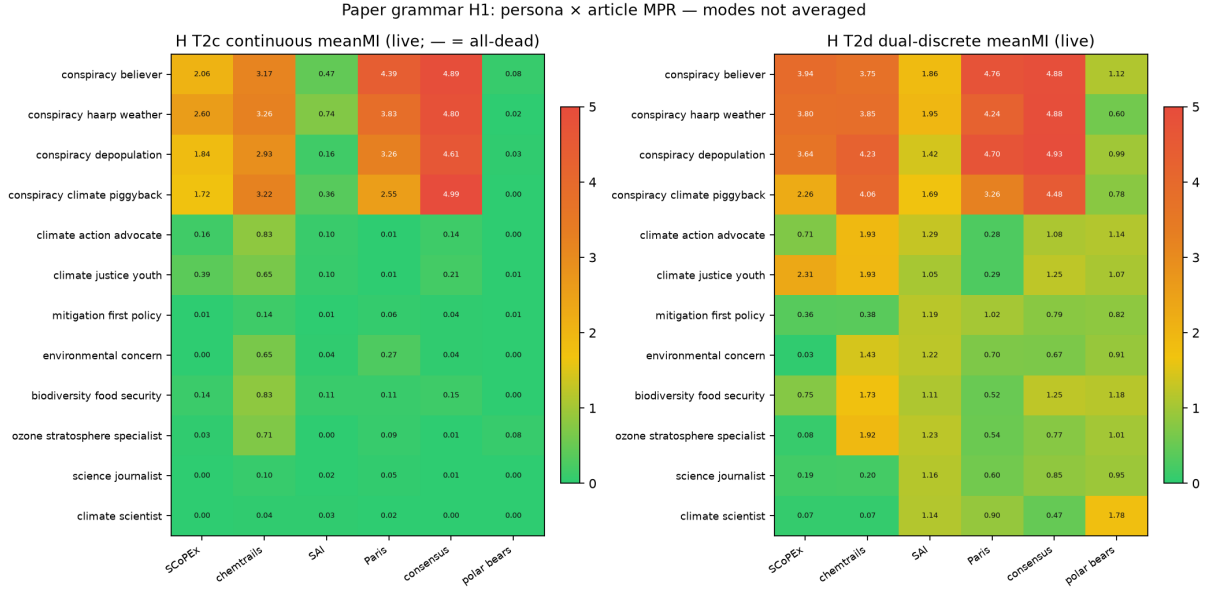


Figure 7: Homogeneous persona $\times$ article heatmaps of live-cell meanMI. Left: continuous. Right: dual-discrete. The two panels are not averaged. Dead cells are not drawn as zero. $N = 1$. Source: `fig10_h_persona_article_mpr.png`.

Dead He cells are not evidence that a mix immunises a network. They are hatched after real LLM calls, as in Figure 5.

5.1 Homogeneous personas

Figures 7 and 9 and Table 5 report the twelve H personas collapsed across topologies. Conspiracy-family cells are the high-MPR group on both instruments (discrete 3.1691, continuous 2.2194). Science/environment is lowest on continuous (0.1118) but not uniformly lowest on dual-discrete (0.8687 versus climate-action 1.0488). Scientist and journalist personas often sit near 0 on continuous and still pick up discrete points on some articles (Figure 10). That disagreement is a mode effect. It is not evidence that “the scientist believed chemtrails”.

Figure 7 is the persona $\times$ article heatmap analogue of the previous paper’s identity-by-story grid. Figure 10 is the matching k^* grid (grey is no irreversible crossing, not zero). Figure 11

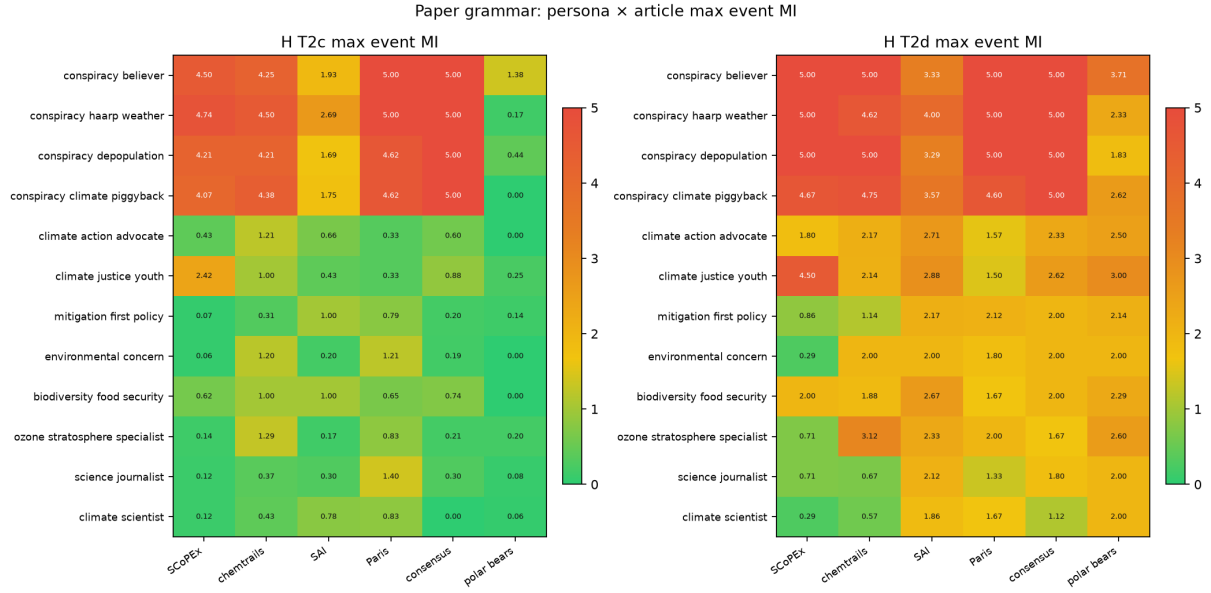


Figure 8: Homogeneous persona $\times$ article heatmaps of live-cell max MI. Left: continuous. Right: dual-discrete. Dead cells are not drawn as zero. $N = 1$. Source: `fig11_h_persona_article_maxmi.png`.

is the persona $\times$ topology grid. Conspiracy rows stay dark on both instruments. Science rows stay pale on continuous and only moderately darker on discrete. Article columns also differ: `polar_bears` is near zero on continuous H, while `climate_consensus` and the `chemtrails`–`Gates` article sit higher. Section 6 returns to the same-mode H/He split as comparison C1.

5.2 Heterogeneous mixes

Figures 12 and 14 and Table 6 report the six He mixes. `mix_00` (four conspiracy + four climate/environment) and `mix_02` (zero conspiracy + eight science/climate) are the useful contrast. Continuous: `mix_00` 1.7712 versus `mix_02` 0.2645. Dual-discrete: `mix_00` 3.1321 versus `mix_02` 1.1882. If `mix_02` is lower, that is “this conspiracy-free mix scored lower in these $N = 1$ runs”, not “diversity always stops firestorms”.

`mix_01` (two conspiracy + six climate/environment) sits between `mix_00` and `mix_02` on both instruments. `mix_03` (one conspiracy + seven other) is enough, in this harvest, to lift dual-discrete MPR to 2.0410 and continuous MPR to 0.8945. `mix_04` and `mix_05` replace core conspiracy identifiers with conspiracy-adjacent extras; they do not form a dose–response law with `mix_00`.

Dual k^* on `mix_00` and `mix_01` exceeds 57%. `mix_02` has dual k^* 0.0% and continuous k^* 0.0%. The expert-inclusive mix never locked network-mean $MI > 3$ in these live cells. That is a descriptive zero on eight ticks, not a proof that experts prevent propaganda.

6 Five comparisons (C1–C5)

The full analysis numbers five contrasts on the live harvest (Maurya et al., 2025). C1 holds scoring mode and varies identity mix. C2 and C3 hold the arm and vary the instrument. C4 crosses both factors and is not a headline. C5 concatenates live cells; that concatenation is

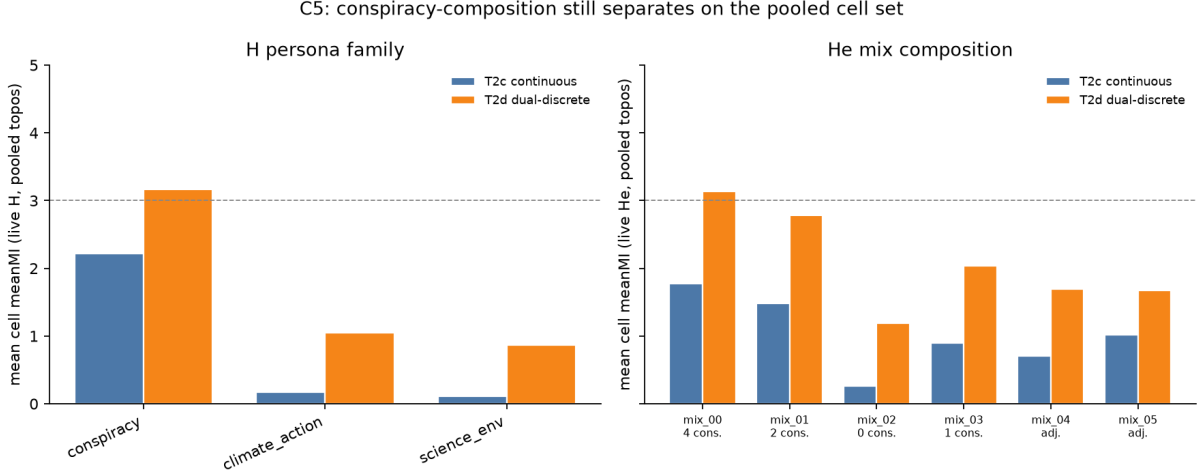


Figure 9: Persona families on homogeneous cells and mix composition on heterogeneous cells. Discrete and continuous remain separate. $N = 1$. Source: `fig07_c5_conspiracy_composition.png`.

not a physical super-graph. Each contrast is descriptive. $N = 1$ supplies point estimates only. Exploratory rank tests in `COMPARISONS.md` are not treated as replicate inference.

6.1 C1: same topology, H versus He, same mode

C1 pairs T2c_H with T2c_He, and T2d_H with T2d_He. Figures 15 and 16 and Table 7 report the topology split.

Cell-pooled same-mode $\Delta(\text{He} - \text{H})$ is 0.1549 continuous (H 0.8643 versus He 1.0192) and 0.3986 dual-discrete (H 1.6814 versus He 2.0800). Topology-equal means do not flip the sign (continuous 0.1725; dual-discrete 0.3974). He exceeds H on 4 of 8 topologies in continuous mode and on 8 of 8 in dual-discrete mode. Continuous $\text{He} > \text{H}$ is therefore not uniform: He is higher on small-world, echo chamber, polarised, and hierarchical graphs; H is higher on linear chain, ring, ER, and scale-free.

That pattern is not “hetero buffers firestorms”. Collapsed H mixes conspiracy personas with scientists. Collapsed He mixes conspiracy-heavy `mix_00/mix_01` with conspiracy-free `mix_02`. Dead He cells remain hatched after LLM calls.

6.2 C2: homogeneous arm, different instruments

C2 holds the homogeneous arm and compares T2c_H with T2d_H. Table 8 reports the topology split. Dual-discrete sits above continuous on 8 of 8 H topologies. The mean topology $\Delta(\text{disc} - \text{cont})$ is 0.8201. Paired live cells (same topology $\times$ persona $\times$ article, both live) number 398; the median paired Δ is 0.8125. Cells dead on one side only were not filled with zero (T2c 80, T2d 82, both dead 16).

This gap is a scoring-mode difference. It is not evidence that the dual campaign found more misinformation in a shared unit. Dual gap on H live cells (discrete minus sidecar continuous on the same dual events) is 0.8804. That sidecar is not the T2c headline.

Table 3: Homogeneous (H) topology headlines. Each topology has 72 cells per mode (12 personas $\times$ 6 articles). Live means exclude hatched dead cells.

Topology	n live dual	Disc. MPR	k^* dual	Dead dual	n live cont.	Cont. MPR	k^* cont.	Dead cont.
linear chain	68/72	1.6127	14.7%	5.6%	62/72	0.9129	14.5%	13.9%
ring	61/72	1.6402	18.0%	15.3%	65/72	0.9160	15.4%	9.7%
random ER	56/72	1.7809	19.6%	22.2%	57/72	0.8736	14.0%	20.8%
small world	59/72	1.5977	20.3%	18.1%	56/72	0.7215	12.5%	22.2%
scale-free	57/72	1.7034	21.1%	20.8%	60/72	0.9205	16.7%	16.7%
echo chamber	61/72	1.7984	23.0%	15.3%	63/72	0.8490	12.3%	12.5%
polarised	57/72	1.7005	21.1%	20.8%	65/72	0.8356	15.4%	9.7%
hierarchical	59/72	1.6315	23.7%	18.1%	52/72	0.8751	11.5%	27.8%

Table 4: Heterogeneous (He) topology headlines. Each topology has 36 cells per mode (6 mixes $\times$ 6 articles).

Topology	n live dual	Disc. MPR	k^* dual	Dead dual	n live cont.	Cont. MPR	k^* cont.	Dead cont.
linear chain	33/36	2.0852	24.2%	8.3%	32/36	0.8638	0.0%	11.1%
ring	33/36	2.1768	39.4%	8.3%	33/36	0.8760	3.0%	8.3%
random ER	29/36	2.0412	27.6%	19.4%	29/36	0.6593	0.0%	19.4%
small world	32/36	1.9096	28.1%	11.1%	31/36	0.7545	3.2%	13.9%
scale-free	29/36	1.9924	31.0%	19.4%	30/36	0.8157	6.7%	16.7%
echo chamber	30/36	1.9632	23.3%	16.7%	32/36	1.3775	9.4%	11.1%
polarised	26/36	1.8830	42.3%	27.8%	25/36	1.4137	12.0%	30.6%
hierarchical	28/36	2.5934	42.9%	22.2%	26/36	1.5242	30.8%	27.8%

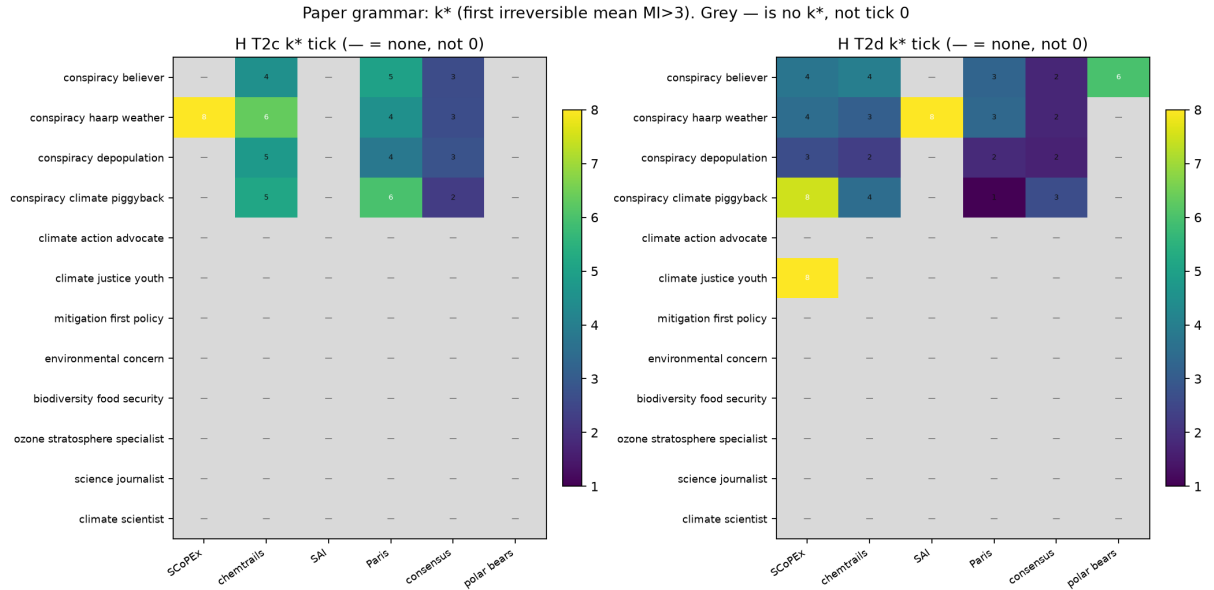


Figure 10: Homogeneous persona $\times$ article k^* . Grey denotes no irreversible crossing, not $k^* = 0$. $N = 1$. Source: fig12_h_persona_article_kstar.png.

Table 5: Homogeneous personas collapsed across topologies. Family labels are analysis groupings, not Debnath HDBSCAN clusters.

Persona	Family	Disc. MPR	Cont. MPR	k^* dual	k^* cont.
conspiracy_believer	conspiracy	3.3522	2.4842	65.8%	47.7%
conspiracy_haarp_weather	conspiracy	3.3571	2.4829	70.5%	47.5%
conspiracy_depopulation	conspiracy	3.2823	1.9535	59.5%	34.9%
conspiracy_climate_piggyback	conspiracy	2.6838	1.9570	42.5%	31.7%
climate_action_advocate	climate action	1.0665	0.2258	0.0%	0.0%
climate_justice_youth	climate action	1.3072	0.2454	2.2%	0.0%
mitigation_first_policy	climate action	0.7571	0.0464	0.0%	0.0%
environmental_concern	science/env.	0.8359	0.1423	0.0%	0.0%
biodiversity_food_security	science/env.	1.1470	0.2115	0.0%	0.0%
ozone_stratosphere_specialist	science/env.	0.9639	0.1627	0.0%	0.0%
science_journalist	science/env.	0.6745	0.0310	0.0%	0.0%
climate_scientist	science/env.	0.7517	0.0130	0.0%	0.0%

Table 6: Heterogeneous mixes collapsed across topologies.

Mix	Composition	Disc. MPR	Cont. MPR	k^* dual	k^* cont.
mix_00	4 conspiracy + 4 climate/env.	3.1321	1.7712	57.5%	12.8%
mix_01	2 conspiracy + 6 climate/env.	2.7795	1.4770	58.5%	9.8%
mix_02	0 conspiracy + 8 science/climate	1.1882	0.2645	0.0%	0.0%
mix_03	1 conspiracy + 7 other	2.0410	0.8945	33.3%	10.3%
mix_04	2 conspiracy-adj. + 6 other	1.6896	0.7021	20.5%	5.3%
mix_05	2 conspiracy-adj. + 6 other	1.6713	1.0134	23.3%	7.5%

6.3 C3: heterogeneous arm, different instruments

C3 holds the heterogeneous arm and compares T2c_He with T2d_He. Table 9 reports the topology split. Dual-discrete sits above continuous on 8 of 8 He topologies. The mean topology $\Delta(\text{disc} - \text{cont})$ is 1.0450. Paired live He cells number 198; the median paired Δ is 0.9778. Dual gap on He live cells is 1.1145; agreement is 0.6174.

Figures 17 and 18 show C2 and C3 together. The two instruments remain separate leagues.

6.4 C4: cross-mode, two factors at once

C4 pairs T2c_H with T2d_He, and T2d_H with T2c_He. The table is labelled and unpooled. No C4 headline is formed by averaging the two cross-mode deltas or by averaging discrete with continuous.

Mean topology $\Delta(\text{T2d_He} - \text{T2c_H})$ is 1.2176. That contrast mixes a higher-scoring instrument with the He composition mix. It is not an identity effect. Mean topology $\Delta(\text{T2c_He} - \text{T2d_H})$ is -0.6476 . Signs reverse because the scoring-mode gap is large. Figure 19 is the reason C4 must stay unpooled.

Article valence is a secondary cut, not a sixth comparison. Figures 20 and 21 show H and He article means on each instrument. Chemtrails exceeds SCoPEX in the article collapse on both arms and both instruments. The contrast still confounds topic, frame, and prior conspiracy fit.

7 C5: cell pooling is not a super-graph

C5 concatenates live article-cells across the eight 8-node topologies. That operation is a table union. It is not a physical 64-node graph, and it is not the 63-node D-net custom topology.

Table 7: C1 live-cell mean `meanMI` by topology. Δ is He minus H inside one instrument.

Topology	T2c H	T2c He	Δ He–H	T2d H	T2d He	Δ He–H
linear chain	0.9129	0.8638	−0.0491	1.6127	2.0852	0.4725
ring	0.9160	0.8760	−0.0400	1.6402	2.1768	0.5366
random ER	0.8736	0.6593	−0.2143	1.7809	2.0412	0.2602
small world	0.7215	0.7545	0.0330	1.5977	1.9096	0.3120
scale-free	0.9205	0.8157	−0.1048	1.7034	1.9924	0.2890
echo chamber	0.8490	1.3775	0.5284	1.7984	1.9632	0.1647
polarised	0.8356	1.4137	0.5780	1.7005	1.8830	0.1825
hierarchical	0.8751	1.5242	0.6490	1.6315	2.5934	0.9618

Table 8: C2 live-cell means. Δ is dual-discrete minus continuous on homogeneous cells. Dual gap is sidecar continuous on the same dual events, not the T2c headline.

Topology	T2c H	T2d H	Δ	n T2c	n T2d	Dual gap
linear chain	0.9129	1.6127	0.6999	62	68	0.9672
ring	0.9160	1.6402	0.7242	65	61	0.8803
random ER	0.8736	1.7809	0.9073	57	56	0.9680
small world	0.7215	1.5977	0.8762	56	59	0.8255
scale-free	0.9205	1.7034	0.7829	60	57	0.8463
echo chamber	0.8490	1.7984	0.9494	63	61	0.8923
polarised	0.8356	1.7005	0.8649	65	57	0.8553
hierarchical	0.8751	1.6315	0.7564	52	59	0.7973

Figure 22 shows the four instrument-by-arm headlines on the concatenated set.

On the concatenated set, He still exceeds H on both instruments (continuous $\Delta = 0.1549$; dual-discrete $\Delta = 0.3986$). Topology-equal weighting does not flip the sign (Tables 11 and 12). The sign is compatible with conspiracy composition, not with a buffering law.

T2c family mean MI on pooled H cells is conspiracy 2.2194, climate action 0.1729, science/environment 0.1118. T2d family means are conspiracy 3.1691, climate action 1.0488, science/environment 0.8687. On pooled He cells, `mix_00` (four conspiracy) is T2c 1.7712 versus `mix_02` (zero conspiracy) 0.2645; T2d `mix_00` is 3.1321 versus `mix_02` 1.1882. Restricting H to the conspiracy family reverses the naive H/He story: conspiracy H sits well above overall He. Figure 23 records that composition split.

Invalid extra pools. Averaging discrete with continuous H still has no auditor ($((0.8643 + 1.6814)/2 = 1.2729)$). Averaging H with He inside one instrument still hides composition. D-net rows are not concatenated into Table 12. Section 8 treats the 63-node custom graph as a separate object.

8 D-net, Pfeffer observables, and Debnath

The D-net arm imports a reconstructed graph and runs the same two scoring modes on homogeneous conspiracy and mixed identities. The empirical file is `thesisExperiment/data/derived/debnath_hashtag_cascade.json`. Kind: `hashtag co-occurrence`. `notARetweetCascade: true`. Nodes are users keyed by published hashtags (63 nodes, 228 directed edges, 171 unique undirected pairs). Edges reuse a `retweets[]` container only as a directed-edge slot. The graph is not a hydrated retweet cascade. The comparable simulated object is this 63-node custom topology, not the eight 8-node T2 generators.

Table 9: C3 live-cell means. Δ is dual-discrete minus continuous on heterogeneous cells. Do not pool T2c with T2d.

Topology	T2c He	T2d He	Δ	n T2c	n T2d	Dual gap
linear chain	0.8638	2.0852	1.2214	32	33	1.2960
ring	0.8760	2.1768	1.3008	33	33	1.3123
random ER	0.6593	2.0412	1.3818	29	29	1.1335
small world	0.7545	1.9096	1.1552	31	32	1.1049
scale-free	0.8157	1.9924	1.1767	30	29	1.1058
echo chamber	1.3775	1.9632	0.5857	32	30	0.8830
polarised	1.4137	1.8830	0.4693	25	26	0.8784
hierarchical	1.5242	2.5934	1.0692	26	28	1.1354

Table 10: C4 live-cell means. The two Δ columns use different instrument pairings.

Topology	T2c H	T2d He	Δ	T2d H	T2c He	Δ
linear chain	0.9129	2.0852	1.1724	1.6127	0.8638	-0.7489
ring	0.9160	2.1768	1.2608	1.6402	0.8760	-0.7642
random ER	0.8736	2.0412	1.1675	1.7809	0.6593	-1.1216
small world	0.7215	1.9096	1.1881	1.5977	0.7545	-0.8432
scale-free	0.9205	1.9924	1.0719	1.7034	0.8157	-0.8877
echo chamber	0.8490	1.9632	1.1141	1.7984	1.3775	-0.4209
polarised	0.8356	1.8830	1.0474	1.7005	1.4137	-0.2868
hierarchical	0.8751	2.5934	1.7182	1.6315	1.5242	-0.1074

Hydration of Debnath’s tweet identifiers failed. Mendeley 10.17632/546hsym93p.1 is an identifier dump, not tweet text (Debnath et al., 2023a). 814,924 tweets were not hydrated. Belief profiles remain theory-faithful reductions, not skip-gram/HDBSCAN centroids (Debnath et al., 2023b; Debnath, 2023). Debnath reports no empirical 0–5 MPR. Simulated auditor MI is therefore not Twitter MI (Ruths and Pfeffer, 2014).

Mapping from hashtag cluster to Debnath type was already in the Dnet configs. No new LLM runs were added for this note. Empirical cluster identity is conspiracy 28, climate action 19, environmental 14, expert 2. Mixed Dnet preserves that type mix (conspiracy share 0.444). Homogeneous Dnet overwrites every seat to `conspiracy_believer` (share 1.0). Exact persona-identifier match `reconstruct $\leftrightarrow$ mixed` is 56 of 63; seven seats stay in the same Debnath type on a coarser library id. Averaged compare `conspiracyShare=0.722` is H+He pooled and is not a third mix.

8.1 C5 D-net auditor scores (not pooled into T2)

Figure 24 and Table 13 list LLM-as-judge headlines on the custom graph. On D-net, homogeneous conspiracy exceeds mixed identity on both instruments and both seeds. Continuous SCoPEx: H 1.7349 versus He 0.9983. Continuous chemtrails: H 3.3378 versus He 2.5068. Dual-discrete SCoPEx: H 4.2537 versus He 2.6880. Dual-discrete chemtrails: H 3.7303 versus He 1.5297.

That ordering matches composition (D-net H is conspiracy-only). It does not match the naive eight-topology collapsed `He>H`, which mixed scientists into H. The H/He label does not travel unchanged onto the 63-node graph. Dual versus continuous still must not be averaged. Chemtrails raises continuous MI versus SCoPEx on the same graph; the dual headline does not preserve that article order on He.

Table 11: C5 concatenated live-cell pools. These rows are table unions, not a 64-node graph and not D-net.

Pool	n live	meanMI	meanNodeMPR	k^* rate
T2c_H all topologies	480	0.8643	0.8630	14.2%
T2c_He all topologies	238	1.0192	1.0140	7.6%
T2d_H all topologies	478	1.6814	1.6839	20.1%
T2d_He all topologies	240	2.0800	2.0737	32.1%
T2c_H conspiracy family	168	2.2194	2.2145	40.5%
T2c_H science/env. family	195	0.1118	0.1130	0.0%
T2c_H climate-action family	117	0.1729	0.1724	0.0%
T2d_H conspiracy family	159	3.1691	3.1799	59.7%
T2c_He mix_00/_01 (conspiracy-heavy)	80	1.6204	1.5921	11.2%
T2c_He mix_02 (conspiracy-free)	41	0.2645	0.2648	0.0%

Table 12: C5: cell-pooled versus topology-equal H/He means. Neither column is a super-graph.

Contrast	Pooled H	Pooled He	Δ	Topo-eq. H	Topo-eq. He	Δ
T2c H vs He	0.8643	1.0192	0.1549	0.8630	1.0356	0.1725
T2d H vs He	1.6814	2.0800	0.3986	1.6832	2.0806	0.3974

8.2 Structural comparison

Figure 25 and Table 14 compare empirical hashtag-graph cascade-shape metrics with the mean of 16 simulated D-net cascades. Structural similarity (share of compared metrics within a 30% band) is 0.6885: breadth matches, depth and structural virality (Goel et al., 2016) do not. Kolmogorov–Smirnov tests use $n_{\text{real}} = 1$ and $n_{\text{sim}} = 16$ and are underpowered. DTFS = 0.2754 with `isValidated=false` (threshold 0.70). That table is cascade shape, not a Pfeffer factor.

Empirical transitivity is 0.315; mean local clustering is 0.515; the chemtrails hub has 33 unique neighbours. D-net copies that graph; it does not grow clusters. On 171 unweighted unique-undirected pairs, the binary conspiracy cut has Newman–Girvan $Q = 0.4297$ under mixed labels. The homogeneous assignment is one community and therefore has $Q = 0$. The six assigned graph clusters instead have $Q = 0.5489$; modularity of a label partition is not a clustering coefficient.

8.3 Pfeffer 2014: seven factors, then a remapping

Pfeffer, Zorbach, and Carley (2014) list seven interrelated factors in the Outlook section:

1. speed and volume of communication;
2. binary choices (like/share/retweet as either–or);
3. network clusters;
4. unrestrained information flow;
5. lack of diversity;
6. cross-media dynamics;
7. network-triggered decision processes.

That list is the primary theory. The thesis six-knob vocabulary is a computational remapping, not a restatement of the 2014 paper. Table 15 splits homogeneous from mixed simulation so that homophily 1.0 is not pooled as a third mix.

Identity-edge homophily on the empirical graph is 0.873 (BP-family 0.838). Mixed Dnet reproduces block echo (identity homophily 0.838). Homogeneous Dnet reports homophily 1.0 because identity was erased. The earlier harvest row that averaged H+He into `edgeHomophily=1`

Table 13: D-net auditor headlines on the hashtag co-occurrence graph. Sidecar continuous on dual runs is not the T2c headline.

Run	Article	Mode	Disc. MPR	Cont. MPR
Dnet_c_H_conspiracy	SCoPEX	continuous	—	1.7349
Dnet_c_H_conspiracy	chemtrails-Gates	continuous	—	3.3378
Dnet_c_He_mixed	SCoPEX	continuous	—	0.9983
Dnet_c_He_mixed	chemtrails-Gates	continuous	—	2.5068
Dnet_d_H_conspiracy	SCoPEX	dual-disc.	4.2537	sidecar 2.2495
Dnet_d_H_conspiracy	chemtrails-Gates	dual-disc.	3.7303	sidecar 3.1251
Dnet_d_He_mixed	SCoPEX	dual-disc.	2.6880	sidecar 1.1302
Dnet_d_He_mixed	chemtrails-Gates	dual-disc.	1.5297	sidecar 1.1497

Table 14: D-net structural metrics. Match = within a 30% band of the empirical value. KS/JS: $n_{\text{real}} = 1$, underpowered.

Metric	Emp.	Sim. mean	Match (30%)	Note
depth	4	2.5625	no	longest path from root
breadth	33	40.9375	yes	max nodes at one level
size	61	53.5	—	informational
struct. virality	2.7	1.7963	no	Goel et al. (2016)
speed	n/a emp.	8 ticks	—	do not convert ticks to hours
mean MI	no emp. MPR	2.5529	do not equate	auditor only

is a homogeneous artefact.

Binary choices (Outlook #2) have no engine row. Scored actions are ternary, with reinterpret about 0.46–0.53. That pattern is the opposite of limited like/share interaction. Cross-media (Outlook #6) remains held: no broadcast agent. Surprise is held drip from `user_chemtrails_hub`. Temporal series are not available empirically; the sim clock is eight ticks, and 61 of 63 nodes are reached in all four primary cells. Outlook #4 (hundreds to thousands of weak ties) is not this 63-node hub of degree 33.

The table is an honesty ledger, not a seven-factor confirmation. Clustering here is an empirical-structure result copied into D-net. Echo is label-dependent. Surprise, temporal acceleration, and cross-media were held or missing empirically.

9 Limitations

The harvest itself marks this campaign as not thesis-grade (`thesisGrade: false`). The following limits are part of the result, not an appendix apology.

$N = 1$. One graph seed (42), one run per config. No error bars, no topology significance tests, no permutation tests. Events on the same node share inbox and trust state; nodes share a graph. Treating scored events as i.i.d. replicates would inflate precision.

Eight hops are not CIKM 30. The hop/tick budget is a logged cost compression versus the CIKM chain length of 30 (Maurya et al., 2025). It is not Debnath’s hop protocol and not an empirical firestorm duration. k^* irreversibility is right-censored at eight ticks.

Table 15: Pfeffer observables on the Debnath hashtag fallback versus simulated D-net, H and He split. Empirical valence is a toxicity/hashtag proxy, not MPR. Simulated MI is an auditor score, not Twitter MI.

Observable	Pfeffer 2014 map	Empirical	Simulated (split)
valence	affective character (definition), not an Outlook number	hashtag prior 0.1097; paper toxicity mean 0.17	auditor MI; chemtrails > SCoPEX on continuous H and He
surprise	#1 speed/volume (shock vs drip)	paper SCoPEX spike; not reconstructed	held drip from chemtrails hub
identity	#5 lack of diversity	conspiracy share 0.444	H 1.0; He 0.444; do not pool to 0.722
clustering	#3 network clusters	transitivity 0.315; local 0.515; graph-cluster $Q = 0.5489$	same topology; conspiracy-cut $Q = 0.4297$ mixed, one-community $Q = 0$
echo	#3 + #4 + parts of #5	identity homophily 0.873	He 0.838; H 1.0 by construction
temporal	#1 / #7	not available (no tweet clock)	8 ticks; 61/63 nodes reached
cross-media	#6	held	held (no engine knob)
binary choices	#2 (no thesis knob)	no like/retweet actions	ternary rewrite, not Schelling binary

Debnath’s eight is a skip-gram window. Debnath et al. (2023b) define a context window of eight *words* for skip-gram embeddings. That hyperparameter is unrelated to this campaign’s eight hops.

Hatched dead cells: 292. Dead cells are kept. They are not zeros and they are not mix immunity. Live-only means drop those cells and therefore describe surviving cascades. Hierarchical and some polarised He slices are particularly incomplete.

Two instruments. Dual-discrete headline $\neq$ continuous headline. Dual gap $\neq$ MPR. Section 7 states that concatenating live cells is not a super-graph and must not be mixed with D-net.

Hashtag graph, not a retweet cascade. The Debnath reconstruct is a hashtag co-occurrence fallback. 814k tweet identifiers were not hydrated (Debnath et al., 2023a). BPs are theory-faithful reductions, not HDBSCAN centroids. No empirical MPR exists on Debnath, so simulated MI cannot “match Twitter misinformation”.

KS/JS with $n_{\text{real}} = 1$. Distributional tests cannot support a digital-twin claim. DTFS failed the 0.70 threshold (0.2754).

Pfeffer 2014 has seven factors. The thesis six-knob list is a remapping (Pfeffer, Zorbach, and Carley, 2014). Cross-media is held (no engine knob). Surprise is held as drip. Binary like/share is not the engine’s ternary action set.

Role-play is not a field community. gpt-4o-mini system prompts can produce prompt-faithful conspiracy language (Park et al., 2023). High conspiracy MPR is compatible with obedience to those prompts. It is not, by itself, evidence of stubborn belief or of chemtrails communities in the wild. False news online does travel farther than true news in observational cascades (Vosoughi, Roy, and Aral, 2018); this simulator does not re-estimate that fact.

Auditor as truth. Five author-written QA items define MI. A different key would move the numbers. Ground truth is the curated scientific record in the article pack, not a moral ranking of personas. Human-eval templates exist in run directories and were not completed as an IRB study.

Phase 1 is a different campaign. The discrete 12×12 chain tables under `thesisExperiment/results/tables/` are not spliced into these headlines. Fact-check injection (Experiment B) and the distortion keyword heuristic (Experiment C) belong to that earlier drop-in set unless a later pass re-runs them under `runs_phase2/`.

Correction literature is out of scope here. Continued-influence and debiasing results (Lewandowsky et al., 2012) motivated Phase 1 Experiment B. Phase 2 does not estimate recovery.

10 Chapter conclusion

Phase 2 completed the eight-topology $\times$ H/He $\times$ dual/continuous grid (1,728 cells, 0 missing) plus four D-net configs. The harvest is a `real-gpt-4o-mini` $N = 1$ pilot with `thesisGrade=false`. Hatched dead cells number 292.

Four headlines stand, and they are not interchangeable. Homogeneous continuous MPR is 0.8643 ($n = 480$ live). Homogeneous dual-discrete MPR is 1.6814 ($n = 478$). Heterogeneous continuous MPR is 1.0192 ($n = 238$). Heterogeneous dual-discrete MPR is 2.0800 ($n = 240$). C2 and C3 show dual-discrete above continuous on every topology. That is a scoring-mode effect.

C1 shows collapsed He above collapsed H on both instruments, but continuous He>H holds on only 4 of 8 topologies. Composition explains the collapse: conspiracy personas and conspiracy-heavy mixes score high; science personas and `mix_02` score low. C4 signs reverse when the instrument is swapped against the arm. C5 concatenates cells; it is not a super-graph.

On the 63-node D-net, homogeneous conspiracy exceeds mixed identity on both seeds and both instruments. That ordering matches composition and does not match the naive eight-topology He>H label. Pfeffer, Zorbach and Carley list seven Outlook factors; this campaign remaps six knobs, splits H from He on echo, and holds cross-media. Simulated MI is not Twitter MPR. DTFS fails. KS tests with one empirical graph are underpowered.

What an examiner can accept is therefore narrow. The engine produces non-zero, persona-sensitive auditor scores on climate seeds. Conspiracy-conditioned nodes sit far above scientist-conditioned nodes on both instruments in this $N = 1$ harvest. Discrete and continuous campaigns must be reported as two leagues. Pooling must be labelled as a table union. Everything else — topology winners, hetero buffering, Twitter matching, seven-factor confirmation — remains outside what these cells can carry.

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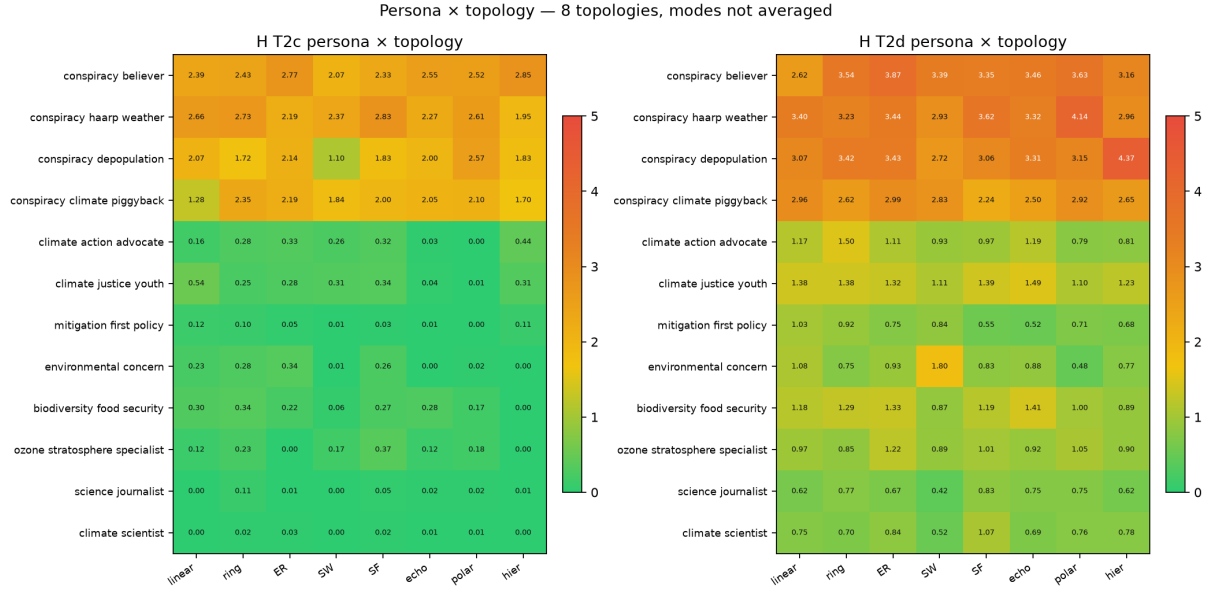


Figure 11: Homogeneous persona × topology heatmaps of live-cell meanMI. Left: continuous. Right: dual-discrete. $N = 1$. Source: `fig15_persona_topology_mpr.png`.

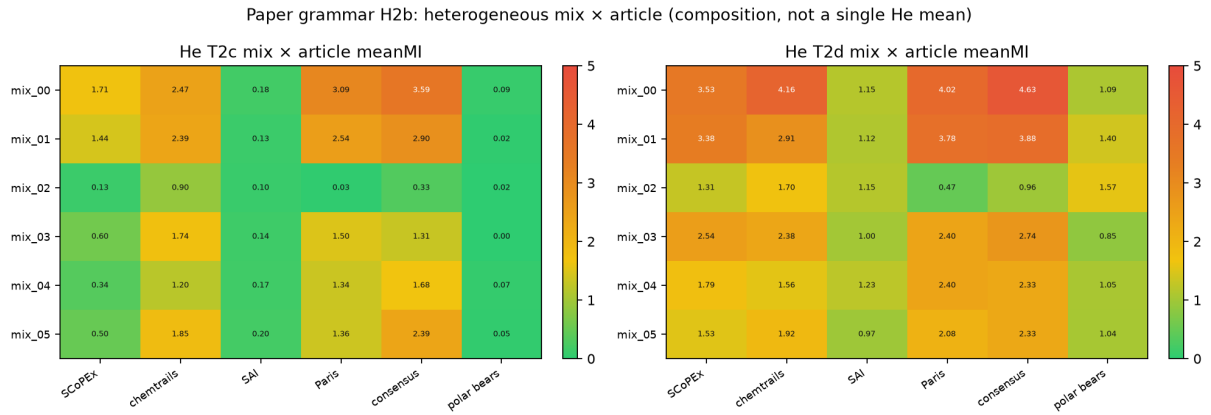


Figure 12: Heterogeneous mix × article heatmaps of live-cell meanMI. Discrete and continuous remain separate. Dead cells are not drawn as zero. $N = 1$. Source: `fig13_he_mix_article_mpr.png`.

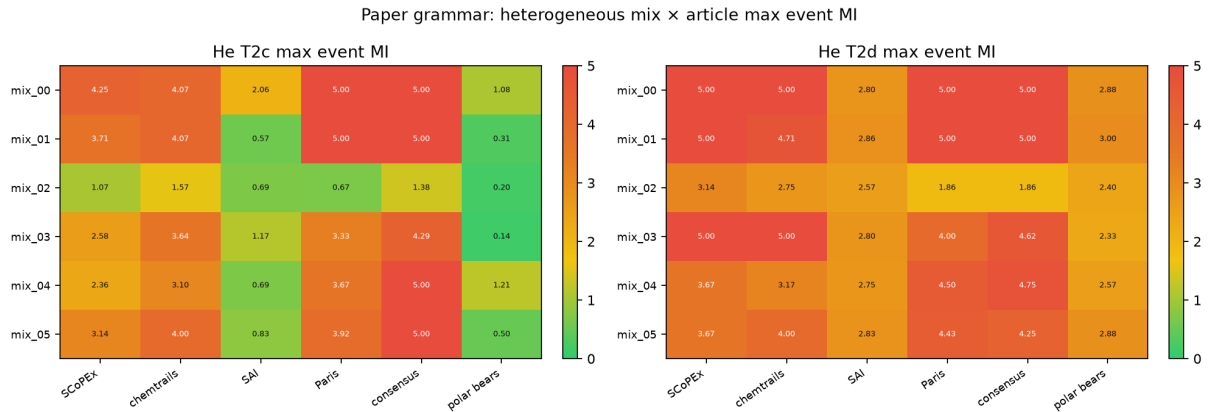


Figure 13: Heterogeneous mix × article heatmaps of live-cell max MI. Discrete and continuous remain separate. Dead cells are not drawn as zero. $N = 1$. Source: `fig14_he_mix_article_maxmi.png`.

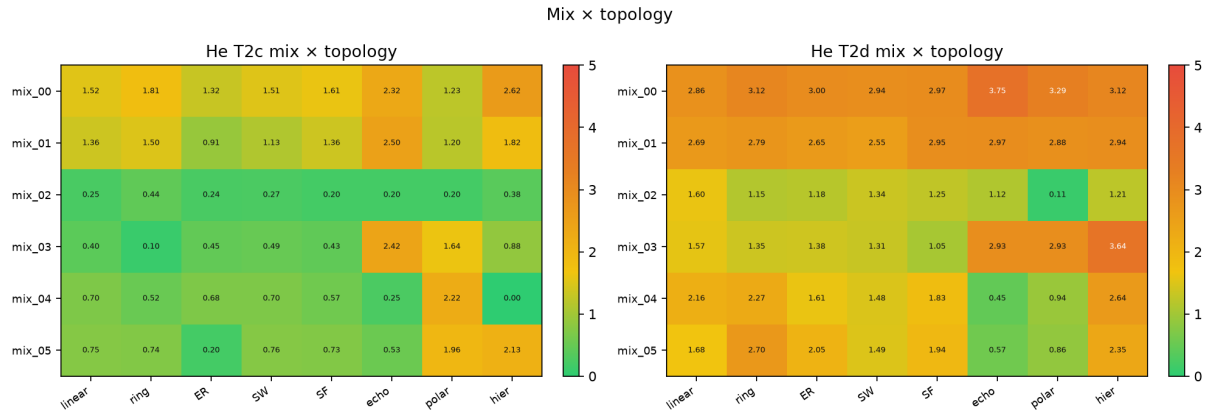


Figure 14: Heterogeneous mix × topology heatmaps of live-cell meanMI. Left: continuous. Right: dual-discrete. $N = 1$. Source: `fig16_mix_topology_mpr.png`.

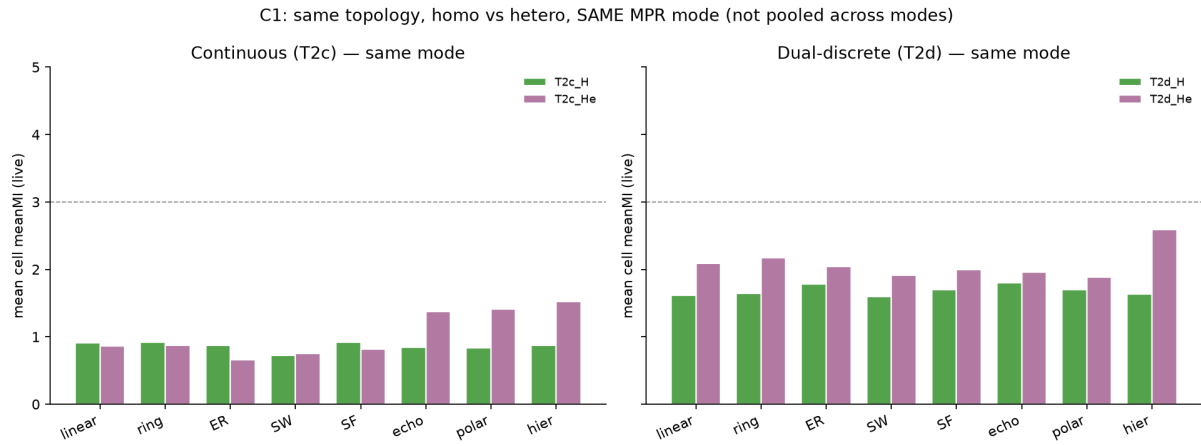


Figure 15: C1: same-mode H versus He by topology. Left: continuous. Right: dual-discrete. The two panels are not averaged. $N = 1$. Source: `fig01_c1_h_vs_he_same_mode.png`.

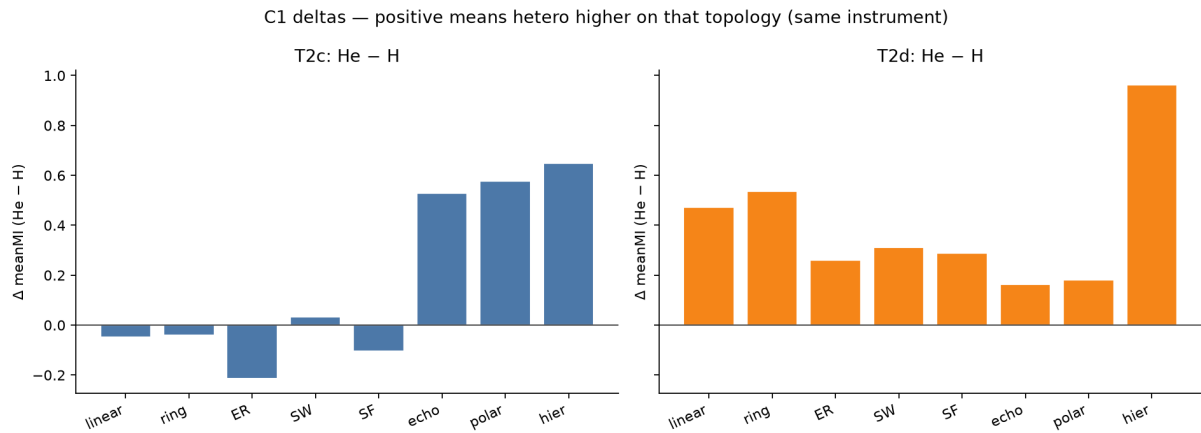


Figure 16: C1: $\Delta(\text{He} - \text{H})$ by topology, separately for each instrument. Positive values mean He scored higher. $N = 1$. Source: `fig02_c1_delta_he_minus_h.png`.

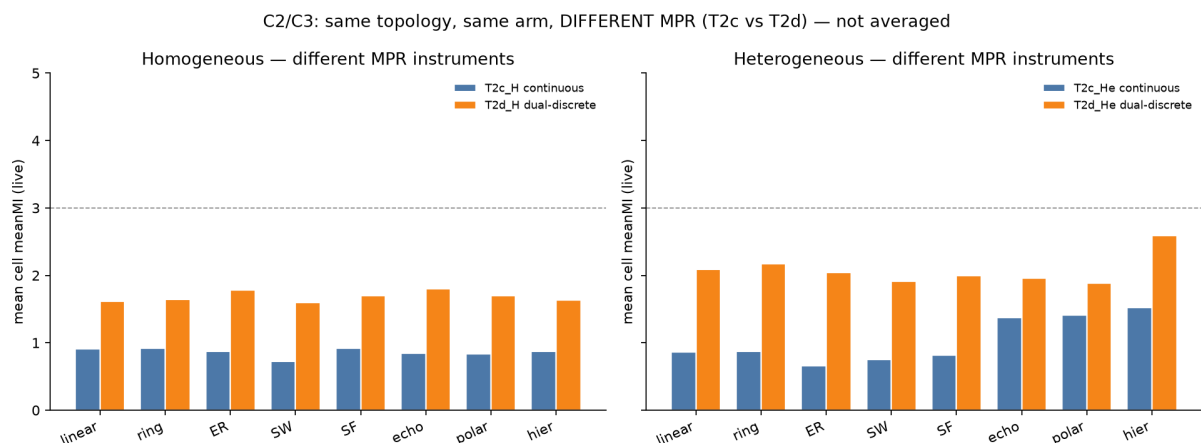


Figure 17: C2 and C3: T2c versus T2d within H and within He. Dual-discrete sits above continuous on every topology in both arms. $N = 1$. Source: `fig03_c2_c3_same_arm_different_mpr.png`.

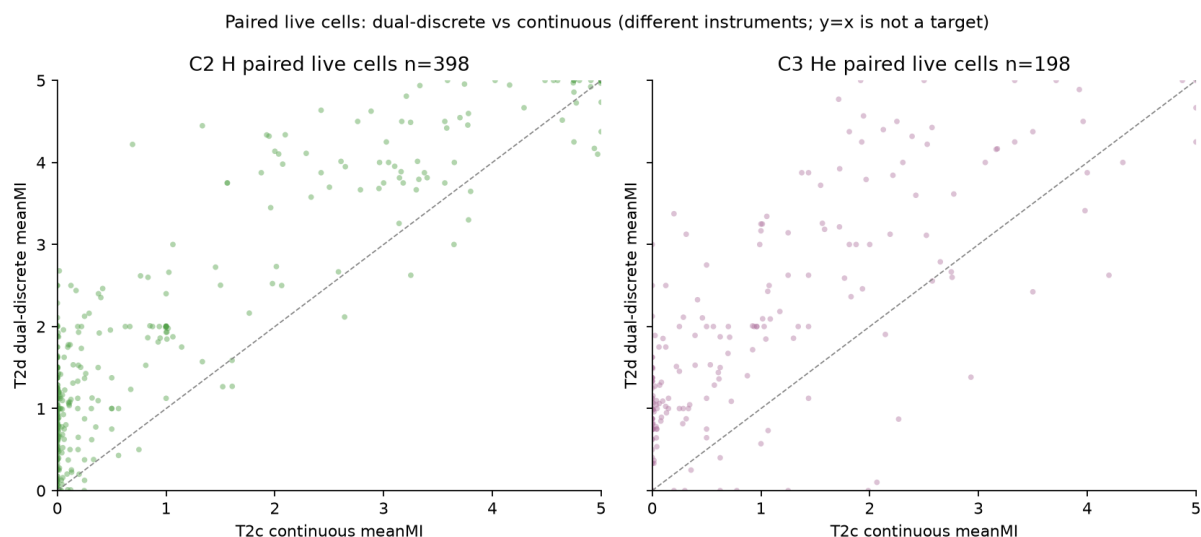


Figure 18: Paired live cells, T2c versus T2d, for homogeneous and heterogeneous arms. Points off the diagonal are instrument disagreement, not a third MPR. $N = 1$. Source: `fig04_c2_c3_paired_scatter.png`.

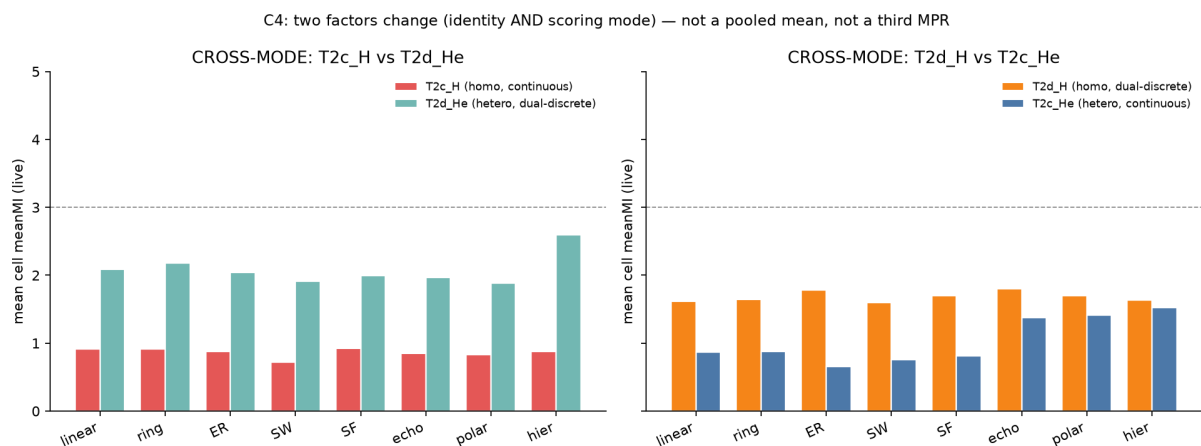


Figure 19: C4: cross-mode two-factor table. The two pairings are not averaged into a third MPR. $N = 1$. Source: `fig05_c4_cross_mode.png`.

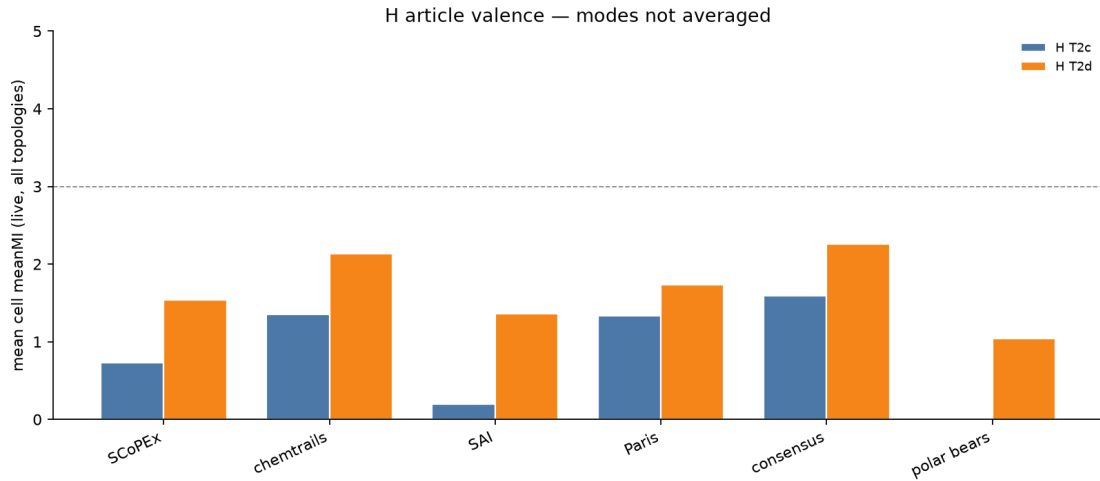


Figure 20: Homogeneous article (valence) collapse, discrete versus continuous. Articles are a factor, not statistical replicates. $N = 1$. Source: fig21_article_valence_h.png.

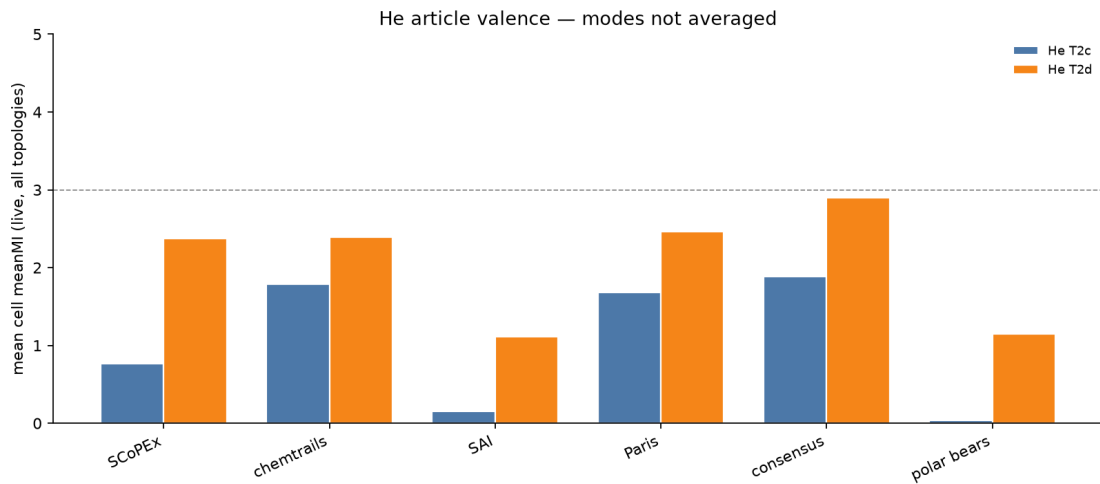


Figure 21: Heterogeneous article (valence) collapse, discrete versus continuous. $N = 1$. Source: fig22_article_valence_he.png.

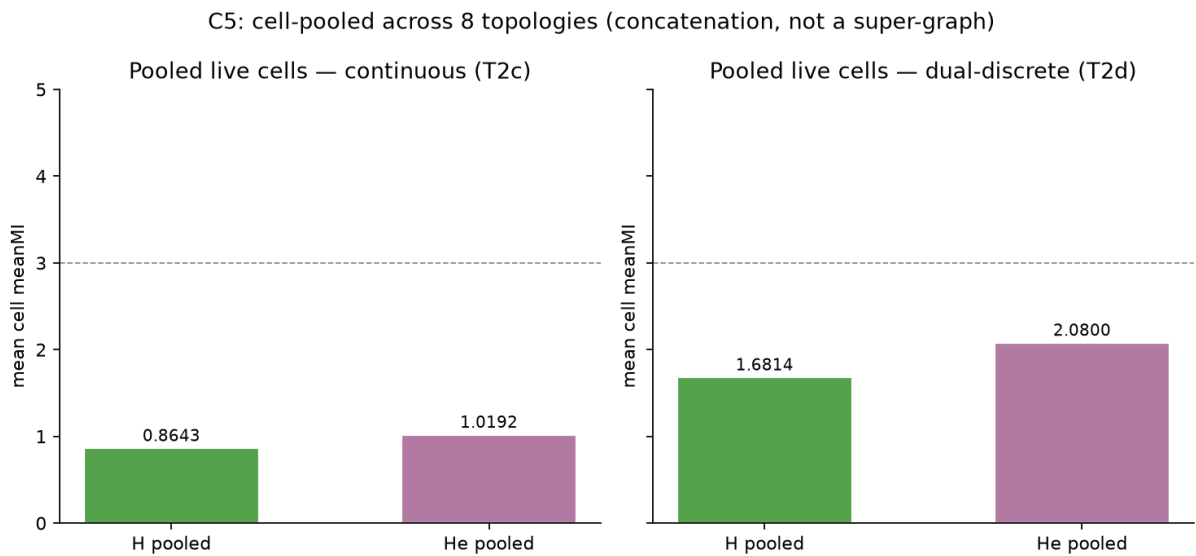


Figure 22: C5: cell-pooled H versus He on concatenated live rows. Pooling is not a super-graph. $N = 1$. Source: fig06_c5_pooled_h_vs_he.png.

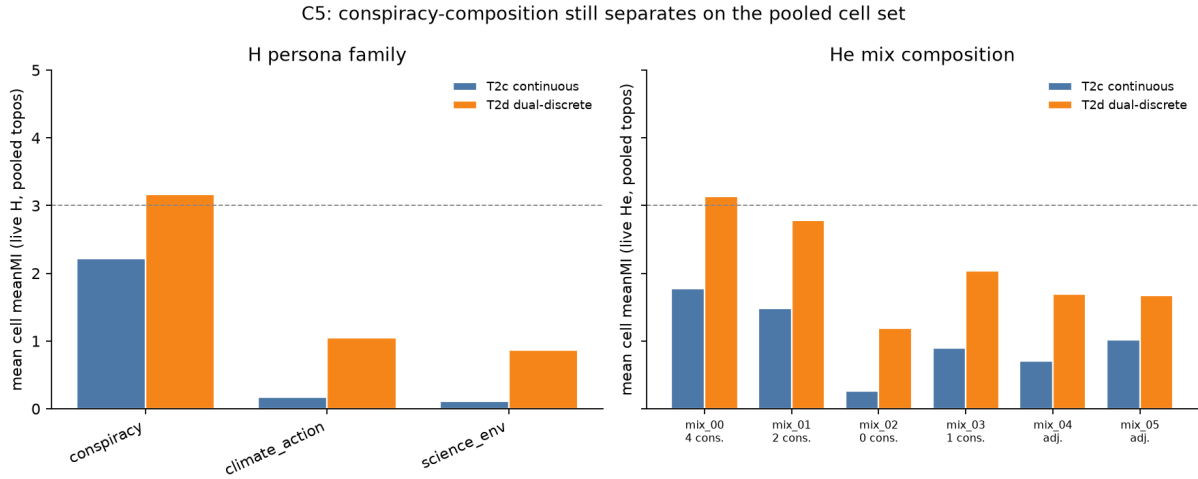


Figure 23: C5: family means on pooled H cells and mix means on pooled He cells. Conspiracy remains the high group on both instruments. $N = 1$. Source: `fig07_c5_conspiracy_composition.png`.

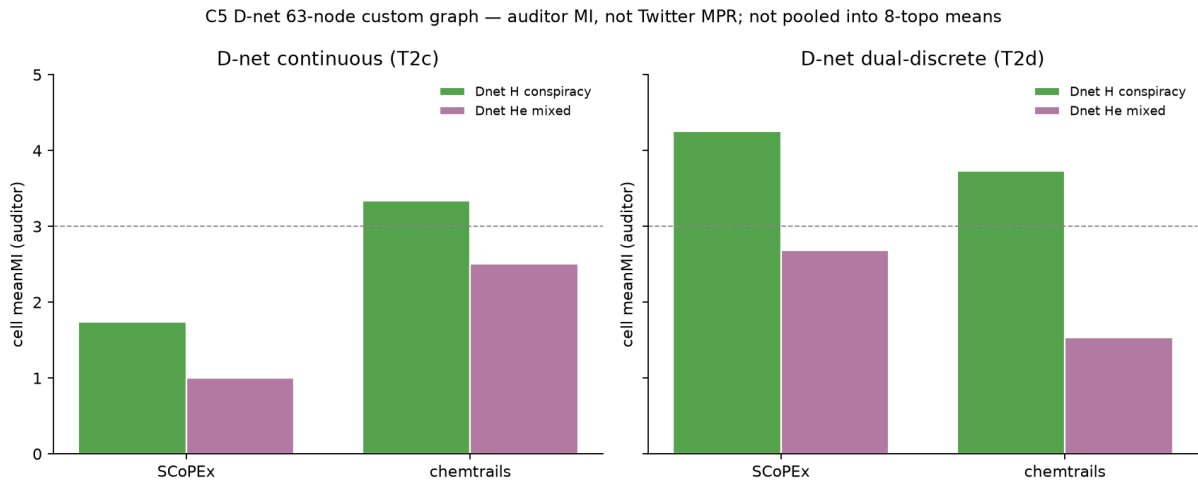


Figure 24: C5: D-net 63-node auditor MI by mix, article, and instrument. Not Twitter MPR. Not concatenated into the eight T2 topologies. $N = 1$. Source: `fig08_c5_dnet.png`.

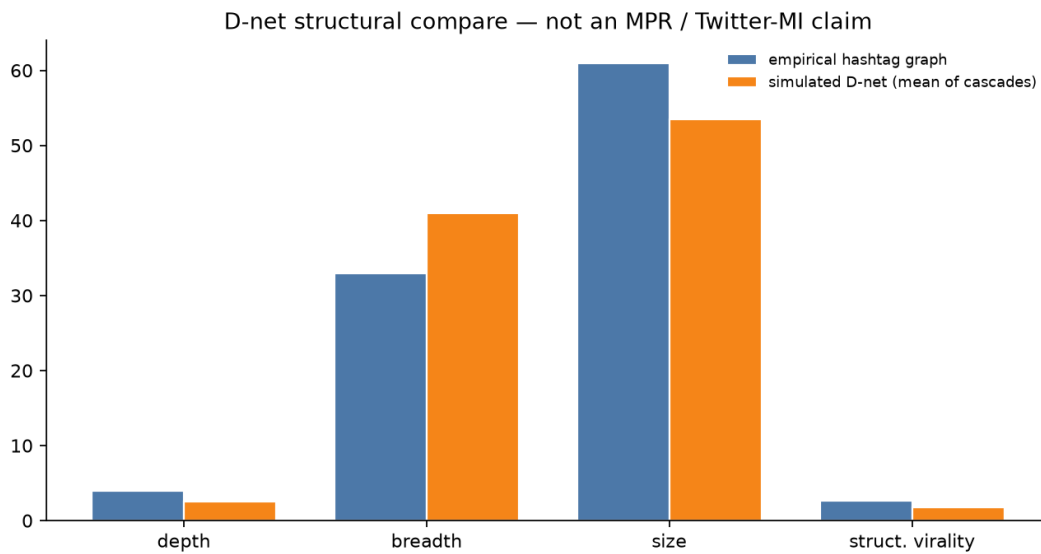


Figure 25: Empirical hashtag-graph metrics versus mean simulated D-net metrics (16 cascades). This is a structural comparison, not an MPR or Twitter-MI claim. Source: `fig09_dnet_structural.png`.